



Technical University of Denmark

Department of Applied Mathematics and Computer Science

02170 Database Systems – **TEST** Examination **Part 1**

Examination date: 6th of May 2026

Course title: Database Systems

Course number: 02170

Aids allowed: All

Exam duration: 3 hours

Weighting: Part 1: 50 %, Part 2: 50 %.

The weighting is only indicative and will ultimately be decided during the evaluation.

This document together with the files [create-database.sql](#) and [answers.sql](#) constitute Part 1 of the written examination.

Part 2 is a digital multiple-choice assignment which should be accessed separately.

Your answers to the questions in this document should be inserted at the right place in the [answers.sql](#) file also provided in this part of the exam. There must **at most be one answer** to each question. Remember to write your student ID at the top of the file.

Feel free to write comments to your answers. Comments must be placed in SQL comments, i.e. just after two hyphens (--) without line breaks.

The provided [create-database.sql](#) file can be used to create and populate the database instance shown in this document. You are welcome to use it while answering the questions. Note that referential integrity constraints are *deliberately omitted* from this file, and they must be added for the database to be consistent, but you are not supposed to do so in this exam.

When you have completed the [answers.sql](#) file, you should **submit it digitally on eksamen.dtu.dk**.

A database for a Company:

In both parts of this examination a database named *CompanyDatabase* is considered. The database contains information about departments, employees, reports, and projects. Below you find the relation schemas of the database and an example of a relation instance of each of these, along with a short description of the schemas.

A *Department* has a unique name (*dname*) and belongs to a single country (*country*). An *Employee* has a unique id (*eid*), a name (*ename*), and is employed in a single department (identified by its *dname*). A *Report* has a unique id (*rid*), a name (*rname*), and a publication year (*ryear*). A *Project* has a unique name (*pname*) and has a single host department (identified by its *dname*).

The *authors* relation contains information about which employees are authoring which reports. The *participates* relation contains information about which employees are participating in which projects. Note that the authors of a report and the participants in a project, respectively, can be employees at different departments.

Department(dname, country)

dname	country
Administration	Denmark
FormalSys	Denmark
Nobody	Utopia
SmartDB	Italy
SmartSys	Germany

Employee(eid, ename, dname)

eid	ename	dname
10	Anne	FormalSys
20	Giovanni	SmartDB
30	Ida	FormalSys
40	Jan	SmartSys
50	Signe	Administration

Report(rid, rname, ryear)

rid	rname	ryear
1	Modern Database Systems	2025
2	Formal Methods in Practice	2024
3	Tools of the Future	2025

authors(rid, eid)

rid	eid
1	10
2	10
2	20
3	10
3	30

Project(pname, dname)

pname	dname
AIfuture	SmartSys
FM	FormalSys
RAISE	FormalSys

participates(pname, eid)

pname	eid
AIfuture	10
AIfuture	20
RAISE	20
RAISE	30

Note: unless explicitly asked for, SQL queries should not return tables having duplicate rows.

Question 1 Write an SQL query, which returns a table containing, for each employee who works for a Danish department and who never authored any report, his/her *ename* and *dname*.

Question 2 Write an SQL query, which returns a table containing, for each report authored by 2 or more employees, the *rid*, *rname*, *ryear*, and total number of employees who authored it. For the number of employees, use attribute name *numAuthors*.

Question 3 Write an SQL query, which returns a table containing, for each project, the *pname*, and the number of reports that were authored by at least one employee participating in that project. For the number of reports, use attribute name *numReports*.

Question 4 Create an SQL function named *reports*, which takes the *eid* (of type INT(3)) of an employee as an argument and returns the number of reports authored by that employee. If the employee *eid* does not exist in *Employee*, it should return 0. The returned number should be of type INT. In the answers.sql file, in the answer to this question, please do not include any DROP FUNCTION command or any SELECT statements using the function.